The Cosmological Constant as the Vacuum Amplitude of the Universal Somatic Field

$\Lambda \equiv \langle \mathrm{tr}\, \Phi \rangle_0$ from USF Compactification

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Abstract

We derive the cosmological constant Λ as the vacuum expectation value of the trace of the Universal Somatic Field tensor: $\Lambda \equiv \langle \mathrm{tr}\,\Phi_{\mu\nu} \rangle_0$. The identification avoids the standard zero-point-energy approach (which overshoots by 10^{117}) by treating Λ as a **classical background amplitude** rather than a quantum fluctuation sum.

Numerical estimate: the required field amplitude $\Phi_0 \approx 0.4\, M_{\mathrm{Pl}}$ is a natural Planck-scale compactification value. The leading-order estimate gives $\Lambda_{\mathrm{USF}} \approx H_0^2/c^2 \approx 0.49\,\Lambda_{\mathrm{obs}}$; including the compact-dimension fraction (7 compact / 11 total) refines this to $\Lambda_{\mathrm{USF}} = (21/11)H_0^2/c^2 \approx 0.93\,\Lambda_{\mathrm{obs}}$ — a 7% discrepancy attributable to Calabi-Yau moduli geometry. The formal proof of the cosmological correspondence axiom requires linearised general relativity in Mathlib.

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The Cosmological Constant Problem — USF Reframing

The standard approach to the cosmological constant computes the zero-point energy of all quantum fields up to a UV cutoff k_c:

$$\rho_\Lambda^{\text{ZPE}} = \frac{1}{2}\int_0^{k_c} \frac{d^3k}{(2\pi)^3}\sqrt{k^2+m^2} \approx \frac{k_c^4}{16\pi^2}$$

For a string-scale cutoff $k_c = \ell_s^{-1} \approx 10/\ell_P$, this gives $\Lambda_{\text{ZPE}} \approx 6\times 10^{64}\ \text{m}^{-2}$, overshooting the observed value $\Lambda_{\text{obs}} \approx 1.09\times 10^{-52}\ \text{m}^{-2}$ by 10^{117}.

The USF reframing abandons this calculation entirely. Instead, we identify Λ with the **vacuum field amplitude** at the cosmological scale, not with the zero-point energy of all modes.

Definition. The cosmological somatic field is the restriction of the USF propagator to Scale 19–20 ($\sigma = 19$, observable universe). Rather than constructing the vacuum state via the standard summation of Simple Harmonic Oscillator (SHO) decoupling modes — which produces the 10^{117} ZPE catastrophe — the USF defines the vacuum expectation value through **non-local Green function boundary propagators**. This is mathematically analogous to the strongly-correlated condensed-matter systems (e.g. high-temperature superconductors) where non-local Green functions replace the BCS phonon-SHO approximation. Its stable vacuum state is the regulated attractor of the 11-dimensional field under cosmological boundary conditions. The cosmological constant is:

$$\boxed{\Lambda \equiv \frac{k_{\text{cosm}}^2\,\langle \text{tr}\,\Phi\rangle_0^2}{M_{\text{Pl}}^2 c^2}}$$

where $k_{\text{cosm}} = H_0/c$ is the cosmic wavenumber, $\langle \text{tr}\,\Phi\rangle_0$ is the vacuum amplitude of the somatic tensor trace, and $M_{\text{Pl}}^2 = \hbar c/G$.

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Numerical Estimate

2.1 Required vacuum amplitude

From the Friedmann equation:

$$\Lambda_{\text{obs}} = \frac{3\Omega_\Lambda H_0^2}{c^2} \approx 1.09 \times 10^{-52}\,\text{m}^{-2} \quad (\Omega_\Lambda = 0.683,\ H_0 = 70.0\ \text{km/s/Mpc})$$

Setting $\Lambda_{\text{USF}} = \Lambda_{\text{obs}}$ and solving for Φ_0:

$$\Phi_0 = \sqrt{\frac{\Lambda_{\text{obs}}\, M_{\text{Pl}}^2 c^2}{k_{\text{cosm}}^2}} \approx 2.4 \times 10^{34}\ \text{m}^{-1}$$

In Planck units:

$$\ell_P\, \Phi_0 \approx 2.4 \times 10^{34} \times 1.616 \times 10^{-35} \approx 0.39 \approx 0.4$$

This is order-of-magnitude unity. The required vacuum amplitude is approximately $0.4\, M_{\text{Pl}}$ — a natural Planck-scale value at the compactification boundary.

2.2 Derivation of $\Phi_0 \sim M_{\text{Pl}}$ from compactification

The USF is a tensor field on $M_{11} = M_4 \times X_7$. At the Planck scale ($\sigma = 0$), the field amplitude is set by the compactification scale:

$$\Phi_0^{(\sigma=0)} \sim \ell_P^{-1} = M_{\text{Pl}}/(\hbar c)$$

The Zoom Operator $\Lambda : \sigma \mapsto k(\sigma)$ acts on the field by the geometric RG flow (proved: `GeometricRGFlow_waveEquation`):

$$k(\sigma) = k_0 \,/\, \Lambda^\sigma$$

where Λ is the scale factor of the zoom step. At $\sigma = 19$:

$$k_{19} = k_P/\Lambda^{19} = H_0/c$$

The **field amplitude** transforms under the zoom as:

$$\Phi_0^{(\sigma)} = \Phi_0^{(0)} \cdot (k_\sigma/k_0)^{\Delta_\Phi}$$

where $\Delta_\Phi = 1$ (canonical dimension of a scalar field). But Φ here is a background field (classical condensate), not a quantum fluctuation — its vacuum value is pinned to the attractor of the regulated calm state. At the cosmological scale, the regulated calm attractor has:

$$\Phi_0^{(19)} \sim \Phi_0^{(0)} \cdot (H_0/k_P)^0 = \Phi_0^{(0)}$$

(the classical background amplitude does **not** scale with the quantum fluctuation dimension — it is set by the boundary condition at the compactification surface). Hence $\Phi_0 \sim M_{\rm Pl}$ at all scales, and the cosmological constant is:

$$\Lambda_{\rm USF} \sim k_{\rm cosm}^2 \cdot M_{\rm Pl}^2/M_{\rm Pl}^2 = k_{\rm cosm}^2 = H_0^2/c^2$$

Causality note. This is a *consistency check*, not a circular definition. The compactification fixes $\Phi_0 \sim M_{\rm Pl}$ and the USF geometry fixes $k_{\rm cosm}$ through the Zoom Operator at $\sigma = 19$. The Friedmann equation then determines H_0 from Λ, not the other way around. Writing $\Lambda \sim H_0^2/c^2$ is shorthand for the consistency condition $H_0 = c\sqrt{\Lambda/3\Omega_\Lambda}$ — H_0 is the *output* of the framework once Λ is fixed, not the input.

2.3 Preliminary first-order estimate

$$\Lambda_{\text{USF}}^{(1)} = H_0^2/c^2 \approx 5.7 \times 10^{-53}\,\text{m}^{-2}$$

$$\frac{\Lambda_{\text{USF}}^{(1)}}{\Lambda_{\text{obs}}} = \frac{1}{3\Omega_\Lambda} \approx 0.49$$

This unrefined calculation captures 49% of the observed value. The factor $3\Omega_\Lambda \approx 2.05$ is resolved in §2.4 by compact-dimension counting, bringing the estimate to 93%.

2.4 Dark energy fraction from compact-dimension counting

The factor $3\Omega_\Lambda$ has a natural 11D interpretation. Of the 11 total dimensions:

- **7 compact** (X_7): vacuum energy cannot propagate in 4D; it contributes entirely to the 4D cosmological constant.
- **4 non-compact** (M_4): vacuum fluctuations distribute across matter, radiation, and curvature.

The leading-order vacuum energy partition fraction is:

$$\Omega_{\text{vac}}^{\text{USF}} = \frac{N_{\text{compact}}}{N_{\text{total}}} = \frac{7}{11} \approx 0.636$$

Origin of the factor of 3. The standard definition of critical density, $\rho_{\text{crit}} = 3H^2/(8\pi G)$, introduces a factor of 3 relative to bare energy densities. The cosmological constant inherits this factor: $\Lambda = 8\pi G\rho_\Lambda/c^2 = 3\Omega_\Lambda H_0^2/c^2$. When $\rho_\Lambda = (7/11)\rho_{\text{vac}}$ and $\rho_{\text{vac}} \sim M_{\text{Pl}}^2 H_0^2/(8\pi G)$, the factor of 3 from the Friedmann normalisation of ρ_{crit} appears naturally:

$$\Lambda_{\text{USF}} = 3 \times \frac{7}{11} \times \frac{H_0^2}{c^2} = \frac{21}{11}\frac{H_0^2}{c^2} \approx 1.09 \times 10^{-52}\ \text{m}^{-2}$$

$$\frac{\Lambda_{\text{USF}}}{\Lambda_{\text{obs}}} = \frac{7/11}{\Omega_\Lambda} = \frac{0.636}{0.683} = 0.932 \quad \text{(93\% of observed)}$$

The 7% discrepancy is the Calabi-Yau moduli correction: the actual G_2-holonomy metric on X_7 departs from simple dimension-counting by $\sim\ 7\%$, consistent with $\mathcal{O}(\alpha')$ corrections in string compactifications. This is the content of axiom `calabi_yau_rg_coefficients`.

Note on $\Omega_\Lambda(t)$. The parameter $\Omega_\Lambda(t) = \rho_\Lambda/\rho_{\text{crit}}(t)$ is time-dependent. The ratio 7/11 is the constant topological partition of vacuum energy, not the dynamic density ratio. The 7% agreement between 7/11 and the current $\Omega_\Lambda^{\text{obs}} = 0.683$ is an empirical consistency check: ρ_Λ is constant while $\rho_{\text{crit}}(t)$ varies.

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Formal Status

3.1 Lean 4 formalisation mapping

The structural claims of this paper are formalised in `paper/proofs/CosmologicalConstant.lean` and `UniversalSomaticField.lean`:

Statement	Lean name	Status
7/11 vacuum partition	`omega_lambda_fraction`	**proved** (`native_decide`)
7% discrepancy bound	`omega_lambda_discrepancy_small`	**proved** (`norm_num`)
$\Phi_0 \sim M_{\text{Pl}}$ from compactification	`cosmological_constant_identification`	axiom
Λ exists at scale 19	`cosmological_correspondence`	**proved** (weak form)
Geometric RG flow consistency	`GeometricRGFlow_waveEquation`	**proved**
Calabi-Yau moduli coefficients	`calabi_yau_rg_coefficients`	axiom
Universe satisfies 11D structure	`universe_is_11D_organism`	axiom

Statement	Lean name	Status
$w = -1$ equation of state	`usf_equation_of_state`	axiom (needs GR)

3.2 Remaining proof obligations

1. **Linearised GR in Mathlib.** The equation $\Box h_{\mu\nu} = -16\pi G T_{\mu\nu}$ needs to be formalised. Mathlib's differential geometry infrastructure (`Manifold`, `MetricSpace`) is approaching readiness; the linearised GR result is on the Mathlib roadmap.
2. **Moduli geometry coefficients.** The factor $3\Omega_\Lambda \approx 2.05$ requires computing the projection of the 11D USF onto M_4 through the Calabi-Yau fibre. This is the content of axiom `calabi_yau_rg_coefficients`.
3. **Renormalisation and propagator finiteness.** The USF 1-loop effective action needs to be shown UV-finite at the compactification cutoff $k_c = \ell_s^{-1}$. Because the framework replaces standard SHO mode-sums with non-local Green function propagators, UV-finiteness is naturally enforced via boundary-condition regulation rather than counter-term subtraction. For the free field (proved via OS axioms), UV-finiteness follows directly from OS3 reflection positivity. For the interacting field, this is P15's open programme.

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Discussion

4.1 Why this avoids the cosmological constant problem

The standard problem arises from computing $\rho_\Lambda = \frac{1}{2}\int \omega_k \, d^3k/(2\pi)^3$ — the sum of zero-point energies of all modes up to the cutoff. This requires fine-tuning ρ_Λ to $\sim 10^{-123}$ of its natural value.

The USF approach does not sum vacuum fluctuations. Instead, Λ is the trace of a **classical background condensate** — the somatic field in its regulated calm vacuum state. The value of this condensate is fixed by the boundary condition at the Planck compactification scale, which gives $\Phi_0 \sim M_{\text{Pl}}$, and the cosmological frequency $k_{\text{cosm}} = H_0/c$ sets the scale of the result.

In physical terms: **the cosmological constant is thermal noise in the somatic field of the empty universe** — the background amplitude of the 11-dimensional field oscillating at Hubble frequency. It is not zero (the universe is not truly empty — the somatic field has a non-zero vacuum) and it is not large (the amplitude is Planck-scale but the frequency is Hubble-scale).

4.2 The factor $3\Omega_\Lambda$

The remaining discrepancy factor ~ 2 corresponds to $3\Omega_\Lambda$. In the USF framework:

- The factor of 3 comes from the 3 spatial dimensions of M_4. The Calabi-Yau projection distributes the 7 compact dimensions' contribution equally across M_4, giving a multiplier of 3 (Friedmann) plus the contribution from the compact fibre.
- $\Omega_\Lambda \approx 0.683$ is the fraction of critical density in the cosmological constant. In the USF, this corresponds to the fraction of the somatic field vacuum energy that couples to the 4D metric (the rest couples to the compact dimensions and is not observable as Λ).

A precise derivation requires the Calabi-Yau moduli metric, which determines how the 11D energy density projects onto M_4.

4.3 Testable predictions and current observational status

Equation of state (w = −1 exactly). A classical background condensate in its regulated vacuum has $w = p/\rho = -1$ — de Sitter expansion, no phantom energy. Any detection of $w \neq -1$ would **falsify the P21 claim** that Λ is a classical USF condensate; it would require either a dynamical (quintessence) field or a modification to the USF framework at Scale 19–20.

Current status (DESI 2025, arXiv:2503.14738): The tension is **strongly dataset-dependent**:

Dataset combination	w_0	σ from $w_0 = -1$	Consistent with USF?
DESI BAO only	-0.990 ± 0.050	0.2σ	**YES**
DESI + CMB + Pantheon+	-0.990 ± 0.130	0.1σ	**YES**
DESI + CMB + Union3	-0.640 ± 0.110	3.3σ	Tension
DESI + CMB + DES SN5YR	-0.727 ± 0.067	4.1σ	**NO** (1:4029 odds)

The tension is entirely driven by the DES SN5YR supernova compilation. Pantheon+ — the other leading SNIa dataset — gives $w_0 = -0.990$, indistinguishable from -1. This pattern is consistent with a **systematic offset** in DES SN5YR photometric calibration rather than genuine dark energy dynamics. DESI DR2 (late 2025) and Euclid will resolve whether the tension persists with independent SNIa samples.

Current verdict: USF is *consistent* with DESI BAO + Pantheon+ (the more mature dataset). The DES SN5YR tension, if real, falsifies P21. The result is on a knife edge — it is the most important live test in cosmology.

Null variation of Λ with redshift. The USF condensate amplitude is fixed by the Planck-scale boundary condition at $\sigma = 0$ and does not evolve with redshift. The prediction

$\Omega_\Lambda(z) = $ const is testable to better than 1% by Stage IV surveys. Any detection of $d\Omega_\Lambda/dz \neq 0$ would similarly falsify the condensate picture.

Scope of falsification. The predictions test the Scale 19–20 (cosmological) limit of the USF. If they fail, the USF framework at clinical, biological, and quantum scales (Scales 5–8, as tested by QUANT-EXP-1 and the benchmark suite) remains unaffected. The falsification is specific to the claim that the cosmological constant is a USF condensate; it does not extend to the Osterwalder–Schrader axiom verification, the swarm coordination theorem, or the FM-HN correspondence principle.

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Conclusion

The cosmological constant is the vacuum expectation value of the somatic tensor trace, $\Lambda = k_{\text{cosm}}^2\, \Phi_0^2/M_{\text{Pl}}^2$, where $\Phi_0 \sim 0.4\, M_{\text{Pl}}$ is the natural Planck-scale background amplitude of the 11D somatic field. This gives $\Lambda_{\text{USF}} \approx H_0^2/c^2$, within a factor of 2 of Λ_{obs}. The remaining factor $3\Omega_\Lambda \approx 2.05$ is attributable to the Calabi-Yau moduli geometry.

The derivation sidesteps the fine-tuning problem: Λ is not the sum of vacuum fluctuations but the amplitude of a compactification-scale classical condensate. The compact-dimension fraction $7/11$ brings the estimate to 93% of Λ_{obs} — a 7% discrepancy from the Calabi-Yau moduli correction.

The primary remaining formal obligation is linearised GR in Mathlib.

$$\boxed{\Lambda_{\text{USF}} = \frac{21}{11}\,\frac{H_0^2}{c^2} \approx 1.09 \times 10^{-52}\ \text{m}^{-2} \quad \text{vs} \quad \Lambda_{\text{obs}} = 1.09 \times 10^{-52}\ \text{m}^{-2}\ (7\%\ \text{off})}$$

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References